EXERCISE 12

Trajectory Crossing and Uniqueness Bounds

Original statement · ProblemsCh2.pdf, p. 3

No crossing • Barrier solutions

PROBLEM STATEMENT

Consider $\dfrac{dy}{dt} = f(y,t)$, where f satisfies the hypothesis of the existence and uniqueness theorem. Suppose that

(i) $y_1(t) = -2$ for all t is a solution and we are studying a different solution for which $y(0) = 0$.
(ii) $y_1(t) = -t - 1$ and $y_2(t) = t^2 + 1$ are solutions and $y(0) = 0$.

Based on the uniqueness theorem, what can you conclude about the solutions in each case?

Source: ProblemsCh2.pdf, Exercise 12 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Hypothesis of the theorem: f and $\partial f/\partial y$ are continuous on an open set $\Omega \subseteq \mathbb{R}^2$ containing the graphs of all the solutions considered. Then through every point $(t_*, y_*) \in \Omega$ passes exactly one solution (on its maximal interval of existence).

Consequence used below (no-crossing principle): if two solutions u, v of this ODE satisfy $u(t_*) = v(t_*)$ at some time t_* in the common interval, then they solve the same IVP at t_*, so $u \equiv v$ on the common interval. Contrapositive: two different solutions never meet. Hence $u - v$ has no zeros on the common interval.

The function f is not given explicitly, and we only use that the theorem applies. All conclusions are therefore restricted to the common interval of existence $J \ni 0$ of the solutions involved (a connected interval containing $t = 0$).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the Intermediate Value Theorem states that a continuous function with no zeros on an interval has constant sign there. Apply it to $u - v$ for two solutions that never meet. The sign at $t = 0$, known from the initial data, then fixes the sign on all of J. This gives bounds without knowing f.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

Picard's theorem of the notes gives one solution per initial value, so two different solutions never meet: if they met at t_1, both would solve the IVP with data at t_1 and would coincide. This is the argument of step 1 of the logistic example of the notes (x never reaches 0 or M, so $M - x$ keeps the sign of $M - x_0$), used here with y_1 and y_2 in place of the constant solutions.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $y_1 \equiv -2$ and a different solution y with $y(0) = 0$

y_1 is a solution and $y \not\equiv y_1$ (indeed $y(0) = 0 \neq -2 = y_1(0)$). Define $d(t) = y(t) - y_1(t) = y(t) + 2$. By the no-crossing principle, $d(t) \neq 0$ for all $t \in J$. d is continuous and $d(0) = 0 + 2 = 2 > 0$. By the Intermediate Value Theorem, a continuous function that changes sign must vanish somewhere between; since d never vanishes, it keeps the sign it has at $t = 0$:

$$y(t) + 2 > 0 \quad \Longleftrightarrow \quad y(t) > -2 \qquad \text{for all } t \in J.$$

Conclusion: the solution starting at $y(0) = 0$ stays strictly above the constant solution $y = -2$ for all time; it is bounded below by -2 and can never reach it.

Part (ii): $y_1 = -t - 1$, $y_2 = t^2 + 1$ and a solution y with $y(0) = 0$

First, check how the two given solutions are placed with respect to each other: $y_2(t) - y_1(t) = t^2 + t + 2$. Its discriminant is $1 - 8 = -7 < 0$, so $t^2 + t + 2 > 0$ for all t (its minimum is $7/4$ at $t = -\frac{1}{2}$). So $y_1 < y_2$ for all t, consistent with the no-crossing principle (they never meet).

Initial position of y: $y_1(0) = -1, y_2(0) = 1$, so $y_1(0) < y(0) = 0 < y_2(0)$. Apply the argument to each pair. $d_1 = y - y_1$ is continuous, $d_1(0) = 0 - (-1) = 1 > 0$ and never vanishes, hence $d_1 > 0$ on J. $d_2 = y_2 - y$ is continuous, $d_2(0) = 1 - 0 = 1 > 0$ and never vanishes, hence $d_2 > 0$ on J. Therefore

$$-t - 1 < y(t) < t^2 + 1 \qquad \text{for all } t \in J.$$

The solution is trapped between the two given solutions for as long as it exists.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Sanity check of the geometry: in (i), the line $y = -2$ and the curve $y(t)$ with $y(0) = 0$ are separated by a vertical gap 2 at $t = 0$; the gap is positive for all t. In (ii), the vertical gap between y_1 and y_2 at $t = 0$ is 2 and $y(0) = 0$ lies strictly inside; at $t = 1$ the bounds are $-2 < y(1) < 2$, and at $t = -1$ they are $0 < y(-1) < 2$.

Limits of the conclusion: it holds only on the common interval of existence J; the bounds alone do not say how large J is. Erratum: if f satisfies the hypotheses of the theorem on the whole region $\mathbb{R}^2$ (continuous with $\partial f/\partial y$ continuous), then in (ii) the solution cannot blow up in finite time, since it is trapped between y_1 and y_2, which are bounded on every bounded interval; hence y is defined for all $t \in \mathbb{R}$ (in (i) there is only a lower bound, so blow-up to $+\infty$ is not excluded). If f is only defined on a smaller region, J may be smaller. If f were not Lipschitz (for example $y' = 3y^{2/3}$, Problem 2.14), different solutions could touch or merge, and the bounds would be lost.

The data in (ii) are a consistency condition for f as well: two solutions that do not meet can coexist, whereas two that cross (say $y_1 = t$, $y_2 = -t$ intersecting at $t = 0$) would contradict the uniqueness hypothesis.

CONCLUSIONS

(i) $y(t) > -2$ for all t in the interval of existence. (ii) $-t - 1 < y(t) < t^2 + 1$ for all t in the interval of existence.